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**"Comparative Analysis of Ablative and Phase-Change Thermal
Protection for Reusable Re-entry Vehicles"**

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Introduction

Problem statement and task formulation

When a spacecraft enters the Earth's atmosphere, its heat shield is exposed to an extreme heat flux reaching 1–10 MW/m². The traditional method of protection is **ablative thermal protection** — the shield material melts, evaporates and burns away, carrying thermal energy with it. This method is reliable but has a significant drawback: the heat shield irreversibly degrades on every descent, requiring complete replacement before the next flight.

An alternative approach is the use of **phase-change materials (PCMs)**, which accumulate thermal energy through melting and solidify upon cooling, retaining their mass and shape. Theoretically, PCMs could provide reusability of thermal protection; however, their use in pure form is limited by the low specific heat of fusion compared to the effective heat of ablation.

The task of this work is to quantitatively compare ablative and phase-change thermal protection by mass and thickness of the thermal protection layer under identical heat loads, and to justify a hybrid scheme combining the advantages of both approaches for reusable re-entry vehicles.

Relevance of the topic

Currently, there is active development of reusable space systems — both in government space agencies (SpaceX, NASA, Roscosmos) and in private companies. One of the key factors in reducing flight costs is the possibility of **multiple reuse** of re-entry vehicles and their components.

Ablative protection used on most modern re-entry vehicles (Soyuz, Crew Dragon, Orion) requires complete replacement of the heat shield after each mission, which increases the cost and time of between-flight preparation. Phase-change materials are able to retain their mass through melt-solidify cycles, making them promising for creating reusable thermal protection systems.

However, in open literature, a systematic comparison of ablative and phase-change materials under identical heat loads is presented only to a limited extent. Most works focus either on improving ablators or on studying PCMs for ground-based heat accumulators. This work fills this gap by offering a simple but illustrative comparison methodology for the early design stages.

Review of analogues

As the baseline ablative protection, the work considers **AVCOAT** — a carbon-phenolic composite used on the Apollo and Orion spacecraft. Its effective heat of ablation is about 20,000 kJ/kg, density — 1400 kg/m³ [1].

Two materials were chosen as PCMs:

1. **Paraffin C24-C50** — an organic phase-change material with a melting point of 45–65 °C, specific heat of fusion of 200 kJ/kg, and density of 800 kg/m³.

Widely used in heat accumulators due to its chemical stability and reversibility of the phase transition [3].

2. **Glauber's salt ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$)** — an inorganic PCM with a melting point of 32 °C, specific heat of fusion of 254 kJ/kg, and density of 1485 kg/m³. It has a higher volumetric energy density compared to paraffin, but is sensitive to corrosion and requires sealing [3].

Research on hybrid thermal protection at MAI. Active research on hybrid thermal protection systems is also conducted at the Moscow Aviation Institute (MAI). MAI scientists develop complex mathematical models of thermal protection composite materials that account for complex physico-chemical processes: binder pyrolysis, gas filtration and phase transitions (melting, sublimation) occurring in materials during aerodynamic heating of hypersonic vehicles [5]. Works from 2024 consider heat and mass transfer in composites with thermal waves caused by phase transitions, which directly resonates with the hybrid scheme proposed in this project [5]. This confirms the relevance of the chosen direction and shows that the proposed solutions are in line with advanced global and Russian scientific developments.

According to numerical simulation data [2], adding a PCM sublayer to the ablative shield reduces the material ablation rate and increases the thermal protection resource, making the hybrid scheme promising for reusable re-entry vehicles.

Environmental aspect

When the ablative layer is destroyed in the atmosphere, ablation products — material particles, melt, gaseous products — are ejected into the upper atmosphere. According to research data, during ablation, melt droplets and microparticles are formed that can deposit on the leeward surfaces of the vehicle or remain in the atmosphere [4].

Although the contribution of re-entry vehicle ablation to the total volume of space debris is difficult to estimate, the growing number of launches makes this problem increasingly relevant. The transition to reusable thermal protection systems with a PCM sublayer potentially reduces the volume of material destroyed and dispersed in the atmosphere, which is consistent with modern trends of sustainable space exploration and reduction of technogenic impact on the upper atmosphere [4].

Main part

2.1. Methods and calculation examples

The following calculation relations were used in the work:

1. **Total thermal energy** acting on the heat shield during descent:

$$Q = q \cdot S \cdot t$$

where Q — total thermal energy (J), q — heat flux (W/m^2), S — frontal surface area (m^2), t — duration of intense heating (s).

2. **PCM layer mass** required to absorb energy Q through melting:

$$m = Q / \lambda$$

where m — PCM layer mass (kg), λ — specific heat of fusion (J/kg).

3. **PCM layer thickness**:

$$h = m / (\rho \cdot S)$$

where h — layer thickness (m), ρ — material density (kg/m^3).

4. **Ablator mass** by effective heat of ablation H_{abl} :

$$m_{\text{abl}} = Q / H_{\text{abl}}$$

5. **Volumetric energy density** for comparative evaluation of materials:

$$E_v = \lambda \cdot \rho$$

where E_v — energy accumulated (or carried away) per unit volume of material (J/m^3).

2.2. Calculation scenario

As a typical case, the descent segment of a small re-entry vehicle (analogue of Soyuz, diameter ~ 2 m) was adopted with the following parameters:

Parameter	Value
Peak heat flux q	1.2 MW/m^2 (convective heating during entry at first cosmo velocity)
Frontal surface area S	3.14 m^2 ($R = 1 \text{ m}$)
Duration of intense heating t	300 s

Calculation of absorbed energy:

$$Q = 1.2 \cdot 10^6 \cdot 3.14 \cdot 300 = 1.13 \cdot 10^9 \text{ J} = 1.13 \text{ GJ}$$

2.3. Obtained data

The calculation results for three materials are summarized in the table.

Parameter	Paraffin C24-C50	Na ₂ SO ₄ ·10H ₂ O	AVCOAT ablator
Specific heat (λ / H_{abl}), kJ/kg	200	254	20,000
Density ρ , kg/m ³	800	1485	1400
Volumetric energy density E_v , MJ/m ³	160	377	28,000
Mass m , kg	5,650	4,450	56.5
Thickness h	2.25 m	0.95 m	12.9 mm
Reusability	Yes	Conditional	No

2.4. Transient thermal calculation

The steady-state calculation presented above gives an upper estimate of the PCM layer thickness, assuming complete melting of the entire volume. In reality, due to the low thermal conductivity of PCM, heat does not have time to penetrate deep into the layer during descent. To verify this effect, a transient thermal calculation was performed using the numerical finite difference method.

Problem formulation. The one-dimensional heat conduction equation with phase transition was solved:

$$\rho \cdot c_{eff}(T) \cdot \partial T / \partial t = \partial / \partial x (k \cdot \partial T / \partial x)$$

where $c_{eff}(T)$ — effective heat capacity accounting for the latent heat of fusion in the temperature range $T_m \pm \Delta T/2$.

Boundary conditions:

- On the surface: heat flux $q = 1.2 \text{ MW/m}^2$ minus radiation according to the Stefan-Boltzmann law
- On the rear side: adiabatic (no heat loss)
- Initial temperature: 20 °C

Calculation parameters:

- Thickness of the modeled layer: 50 mm
- Number of grid nodes: 501
- Time step: $\Delta t \approx 0.04$ s (determined from the stability condition of the explicit scheme)
- Calculation time: 300 s

The calculation results are summarized in the table:

Parameter	Paraffin C24-C50	Na ₂ SO ₄
Surface temperature (300 s)	~1900 °C	~1850 °C
Melting depth	~15 mm	~17.5 mm
Steady-state thickness	2,250 mm	950 mm
Ratio steady/transient	150 times	54 times

Graphical interpretation. The figure shows three graphs:

a) Temperature profile through the layer thickness at time instants $t = 0, 30, 60, 120, 180, 300$ s. It can be seen that the temperature drops sharply within the first 10–15 mm: heat does not penetrate deep.

b) Melting depth as a function of time. In 300 s, only the upper layer of ~15–17.5 mm thickness will melt.

c) Surface temperature. The surface heats up to ~1900 °C within the first seconds and reaches equilibrium (heat input = radiation).

Physical interpretation. The results show that:

- The equilibrium surface temperature $T = (q/\epsilon\sigma)^{1/4} \approx 1927$ °C — matches the calculation;
- The heat penetration depth $L \approx \sqrt{(\alpha t)}$ for paraffin is ~6 mm, for salt ~9 mm — matches in order of magnitude with the results;
- In 300 s, only 0.5–2% of the steady-state layer thickness will melt.

2.5. Notes

Note on the calculation: the presented estimate gives an upper limit of PCM mass and thickness, as it assumes complete melting of the entire layer. In reality, due to the low thermal conductivity of PCM (for paraffin ~0.2 W/(m·K)), in 300 s only a near-surface layer of the order of a few millimeters will melt; the rest of the volume works as thermal insulation. For accurate design, a transient thermal calculation is required.

The transient calculation showed that the PCM surface heats up to ~1900 °C — this is above the decomposition temperatures of paraffin (300–400 °C)

and the loss of crystallization water from Glauber's salt (100 °C). This means that in reality the PCM surface will be destroyed long before equilibrium is reached. This effect is not accounted for in the model and confirms the necessity of an external ablative layer that takes the main thermal load.

Note on the work: the numerical estimates given in the work are approximate, evaluative in nature. This is due to the fact that the initial data for a Soyuz-type re-entry vehicle in different sources and for different modifications (7K-OK, 7K-T, TMA, MS) can differ noticeably.

In particular:

- **Diameter and frontal surface area.** The work adopted a diameter of ~2 m and an area of 3.14 m². The real diameter of the Soyuz re-entry vehicle is about 2.2 m, which gives an area of ~3.80 m². The difference is about 20%. However, this does not change the qualitative conclusions, since the thermal protection layer thickness does not depend on the area, and the mass grows proportionally to the area.
- **Heat flux.** The value of 1.2 MW/m² is a characteristic peak level for the stagnation point during entry at first cosmic velocity. The real heat flux varies with time and over the vehicle surface, so using a constant average value is a simplification.
- **Duration of heating.** The adopted 300 s is a typical estimate of the time of intense aerodynamic heating, but the actual duration depends on the descent trajectory.
- **Material properties.** The values of specific heat of fusion and effective heat of ablation for paraffin, Glauber's salt and AVCOAT are taken from reference literature and may vary depending on the purity of the substance, composite composition, and measurement methodology.
- **Soyuz thermal protection.** The work considers AVCOAT (carbon-phenolic composite) as the ablator. The real Soyuz uses ablative protection based on glass fibers and phenol-formaldehyde binder. These materials are close in class, but not identical; their effective heat of ablation may differ by 10–20%.

Thus, the given values of mass and thickness should be considered as an upper estimate obtained under the assumption of complete melting of the PCM layer and constant heat flux. Refinement of the initial data (area, heat flux, material properties) may change the absolute figures by 20–30%, but does not affect the fundamental conclusions of the work:

1. A pure PCM layer of the required thickness (0.95–2.25 m) is physically impossible to fit within the dimensions of a re-entry vehicle.
2. Ablative protection remains the most mass-efficient, but single-use.
3. The hybrid scheme (ablator + PCM sublayer) is a compromise solution that reduces the volume of replaceable material and environmental load.

For accurate design, a transient thermal calculation is required, accounting for thermal conductivity, heat capacity, and temperature dependence of material properties, which is beyond the scope of this evaluative work.

2.6. Analysis of results

Comparison of the obtained data allows the following conclusions:

1. **By specific energy:** AVCOAT ablator has an effective heat of ablation of 20,000 kJ/kg, which is 80–100 times higher than the specific heat of fusion of PCM (200–254 kJ/kg). This explains the fundamental difference in mechanisms: ablation **carries away** heat with the destroyed material, while PCM only **stores** it through the phase transition.
2. **By mass:** To absorb 1.13 GJ of thermal energy, 56.5 kg of ablator is required, whereas the PCM layer would have a mass of 4.45–5.65 t — 80–100 times more. From an engineering standpoint, this makes pure PCM unsuitable for use as the sole thermal protection layer on a re-entry vehicle.
3. **By thickness:** An ablator 12.9 mm thick is realistic for a 2 m diameter vehicle (Soyuz has an ablative layer thickness of ~10–20 mm). A PCM layer of Glauber's salt would require a thickness of 0.95 m, and paraffin — 2.25 m, which is structurally unfeasible.
4. **By reusability:** The ablator irreversibly degrades and requires complete replacement after each descent. PCMs can theoretically be used multiple times, provided a melt–solidify cycle is organized. However, for Glauber's salt, post-landing cooling below 32 °C and sealing against corrosion must be ensured.
5. **Comparison of PCMs:** Glauber's salt is preferable to paraffin: its volumetric energy density E_v is 2.4 times higher, which allows reducing the layer thickness by 2.4 times at the same heat load.
6. **Environmental factor:** Ablative protection on every descent ejects material destruction products into the atmosphere — melt droplets and microparticles [3]. The hybrid scheme, by reducing the volume of burning ablator by 30–50%, also reduces the amount of ejected material, which is an additional argument in its favor under growing launch numbers and tightening requirements for the protection of the space environment [4].
7. **By the transient calculation:** Numerical modeling showed that in 300 s only 15–17.5 mm of PCM will melt — 54–150 times less than the steady-state estimate. The surface temperature reaches ~1900 °C.

This means that pure PCM fundamentally cannot work as the sole thermal protection: the surface will be destroyed due to material decomposition long before the entire volume is heated.

2.7. Economic assessment of the hybrid scheme

The physical advantages of the hybrid scheme (reduction of ablator thickness by 30–50%) make sense only if economically efficient. To verify this, an evaluative economic calculation was performed.

Initial data and assumptions:

Parameter	Value	Source / justification
Currency exchange rate	84.20 RUB/USD	Central Bank of Russia, 21.09.2

Parameter	Value	Source / justification
Ablative material cost	143,136 RUB/kg	Estimate by analogue (PICA-X), bound
PCM cost (salt + barrier)	~500 RUB/kg	Wholesale price
Cost of launching 1 kg	505,185 RUB/kg	Estimate for Soyuz-2
Ablator replacement (pure)	\$4,000,000	Inter-flight maintenance estimate
Ablator replacement (hybrid, –40%)	\$2,400,000	Proportional to mass
PCM maintenance	\$200,000	Sealing, cooling

Cost comparison over 10 flights:

Item	Pure ablator	Hybrid scheme
Ablator material	80.9 million RUB	48.5 million RUB
PCM material	—	1.1 million RUB
Maintenance (9 cycles)	3,031.1 million RUB	1,970.2 million RUB
Savings on launch mass	—	–1,049.8 million RUB
Total	3,112.0 million RUB	2,019.8 million RUB

Result: even under conservative assumptions (underestimated AVCOAT cost, additional PCM mass ~2 t), the hybrid scheme is economically more advantageous by ~1.1 billion RUB over 10 flights.

The main source of benefit is the reduction of costs for replacing the ablative shield after each flight. The losses on launching the additional PCM mass are covered by this saving.

Calculation limitations:

- AVCOAT cost is taken by analogue (PICA-X) — the real one may be higher
- PCM sublayer mass is estimated approximately (~2 t)
- PCM maintenance cost requires clarification

- The calculation does not account for inflation and changes in launch prices

2.8. Impact of PCM mass on the parachute system

Adding a PCM sublayer increases the mass of the re-entry vehicle, which requires recalculation of the parachute system. At steady-state descent, the drag force of the parachute balances the force of gravity:

$$\frac{1}{2} \cdot \rho \cdot v^2 \cdot C_d \cdot S = m \cdot g$$

Hence the canopy diameter:

$$D = \sqrt{(8 \cdot m \cdot g) / (\pi \cdot \rho \cdot v^2 \cdot C_d)}$$

Key conclusion: the canopy diameter is proportional to the square root of the mass:

$$D \propto \sqrt{m}$$

Numerical example (for a Soyuz-type vehicle, $m_1 = 2900$ kg, $v = 7$ m/s, $\rho = 0.7$ kg/m³, $C_d = 1.5$):

Parameter	Pure vehicle	With PCM sublayer (+2080 kg)	Change
Mass	2,900 kg	4,980 kg	+72%
Canopy area	1,105 m ²	1,898 m ²	+72%
Canopy diameter	37.5 m	49.2 m	+31%
Parachute mass	~100 kg	~170 kg	+70%

Consequences:

1. The parachute diameter grows by 31% — this requires complete redesign of the parachute system.
2. The canopy area grows by 72% → the fabric mass, container volume and deployment loads grow proportionally.
3. The parachute mass increases by another ~70 kg, creating positive feedback: the heavier the vehicle, the larger the parachute, the heavier the vehicle.
4. The parachute system is the hidden cost of the hybrid that was not accounted for in the initial mass calculation.

The increase in parachute mass again increases the vehicle mass, which requires an even larger parachute. Taking this into account, the total mass grows by another ~5–10%.

Conclusion: the parachute system is the hidden cost of the hybrid. It adds ~500 thousand RUB to the cost and ~70 kg to the mass, which reduces but does not outweigh the economic benefit from reducing the ablator (~1.1 billion RUB over 10 flights).

2.9. Expanding horizons: thermal protection for other worlds

All previous calculations were performed for descent to Earth — in an atmosphere consisting of nitrogen and oxygen, at an entry speed of about 7.5 km/s and a peak heat flux of 1.2 MW/m². However, the developed methodology — comparing ablative and phase-change thermal protection — is not tied to a single planet. It is based on fundamental physical principles: the law of conservation of energy, the heat conduction equation and the Stefan-Boltzmann law. This means it can be adapted for any planet with an atmosphere.

The question of thermal protection for other planets is not an abstract task for the future. In 2021, NASA successfully landed the Perseverance rover on Mars using an ablative shield made of PICA. In 2005, the Huygens probe landed on Titan — a moon of Saturn — with an ablative shield made of AQ60. These missions proved that landing on other bodies of the Solar System is technically possible.

Moreover, Mars and Titan are the two most likely candidates for future crewed missions. Mars is close to Earth, has water ice, and gravity sufficient for human adaptation. Titan possesses a dense nitrogen atmosphere protecting against radiation and reserves of organic compounds — potential raw materials for producing fuel and materials.

The question arises: can the hybrid scheme "ablator + PCM sublayer", justified for Earth, be applied to the conditions of Mars and Titan? The answer requires separate analysis, because the atmospheres of these planets differ fundamentally from Earth's. This section examines how the calculations will change, what materials will be required, and where the hybrid scheme works and where it does not.

2.9.1. Universal principle of the hybrid scheme

The hybrid scheme "ablator + PCM sublayer", justified for Earth, is a universal principle that can be adapted for other planets. However, the specific implementation requires complete recalculation for the conditions of each atmosphere.

The basic formula remains:

$$Q = q \cdot S \cdot t$$

But now the heat flux q is the sum of two components:

$$q_{\text{total}} = q_{\text{conv}} + q_{\text{rad}}$$

- q_{conv} — convective heating (depends on density and velocity)
- q_{rad} — radiative heating (depends on gas chemistry and shock layer temperature)

Key rule:

- If q_{rad} is small (as on Earth, Titan) → the hybrid scheme works almost without changes.
- If q_{rad} is large (Mars) → new materials and 3D-CFD modeling are required.

2.9.2. Mars: radiative heating and non-equilibrium chemistry

Atmosphere parameters

Parameter	Value
Composition	95.3% CO ₂ , 2.7% N ₂ , 1.6% Ar
Surface pressure	~6 mbar (0.6% of Earth's)
Density	~0.02 kg/m ³ (60 times less than Earth's)
Average temperature	−63 °C

Entry parameters

Parameter	Value
Entry speed	6–7 km/s (for MSL — 5.8 km/s)
Peak heat flux	40–120 W/cm ² (4–12 MW/m ²)
Heating duration	~200–400 s
Radiative heating share	up to 50–70% of total

Key problem: radiative heating

When entering the CO₂ atmosphere, the shock wave heats the gas to temperatures at which the radiative heat flux becomes comparable to the

convective one. On Mars, this is especially important because on the leeward side radiative heating exceeds convective.

Calculation for Mars (adaptation of the hybrid scheme)

Initial data:

- $q_{\text{total}} = 100 \text{ W/cm}^2 = 1.0 \text{ MW/m}^2$ (peak)
- $t = 300 \text{ s}$
- $S = 3.14 \text{ m}^2$

Absorbed energy:

$$Q = 1.0 \cdot 10^6 \cdot 3.14 \cdot 300 = 0.94 \text{ GJ}$$

PCM thickness (Glauber's salt):

$$h = Q / (\lambda \cdot \rho \cdot S) = (0.94 \cdot 10^9) / (254,000 \cdot 1485 \cdot 3.14) \approx 0.79 \text{ m}$$

Ablator thickness (PICA):

PICA has an effective heat of ablation of $\sim 20,000 \text{ kJ/kg}$ (like AVCOAT), density 270 kg/m^3 .

$$m_{\text{abl}} = Q / H_{\text{abl}} = (0.94 \cdot 10^9) / (20 \cdot 10^6) = 47 \text{ kg}$$

$$h_{\text{abl}} = m_{\text{abl}} / (\rho_{\text{abl}} \cdot S) = 47 / (270 \cdot 3.14) \approx 55 \text{ mm}$$

Conclusion for Mars: PICA ablator thickness is $\sim 55 \text{ mm}$ (more than on Earth due to higher energy). PCM sublayer thickness is $\sim 0.8 \text{ m}$, which is still structurally unfeasible.

Thermal protection material for Mars

For Mars, **PICA** (Phenolic Impregnated Carbon Ablator) is used. PICA was used on MSL, Mars 2020, Stardust. Its density is $0.22\text{--}0.27 \text{ g/cm}^3$, thermal conductivity — $0.22 \text{ W/(m}\cdot\text{K)}$.

Advantage of PICA: high efficiency at low density, which is critically important for Mars.

2.9.3. Titan: "the second home" of humanity

Atmosphere parameters

Parameter	Value
Composition	$\sim 95\% \text{ N}_2$, $1.4\text{--}4.9\% \text{ CH}_4$
Surface pressure	1.5 atm (1.5 times Earth's)

Parameter	Value
Density	~5.3 kg/m ³ (4 times Earth's)
Surface temperature	–179 °C

Entry parameters

Parameter	Value
Entry speed	~6–7.3 km/s
Peak heat flux	~250–300 W/cm ² (25–30 MW/m ²)
Heating duration	~200–300 s
Radiative heating share	80–95% (mainly from CN)

Key problem: radiative heating from CN

When entering the N₂/CH₄ atmosphere, cyanogen (CN) is formed, which emits very strongly in the violet region of the spectrum. Radiative heating accounts for 80–95% of the total.

Calculation for Titan (adaptation of the hybrid scheme)

Initial data:

- $q_{\text{total}} = 250 \text{ W/cm}^2 = 2.5 \text{ MW/m}^2$
- $t = 250 \text{ s}$
- $S = 3.14 \text{ m}^2$

Absorbed energy:

$$Q = 2.5 \cdot 10^6 \cdot 3.14 \cdot 250 = 1.96 \text{ GJ}$$

PCM thickness (Glauber's salt):

$$h = Q / (\lambda \cdot \rho \cdot S) = (1.96 \cdot 10^9) / (254,000 \cdot 1485 \cdot 3.14) \approx 1.65 \text{ m}$$

Ablator thickness (AQ60):

The Huygens probe used AQ60 — a phenolic ablator based on silica fibers. Its effective heat of ablation is ~15,000 kJ/kg, density ~1400 kg/m³.

$$m_{\text{abl}} = Q / H_{\text{abl}} = (1.96 \cdot 10^9) / (15 \cdot 10^6) = 131 \text{ kg}$$

$$h_{\text{abl}} = m_{\text{abl}} / (\rho_{\text{abl}} \cdot S) = 131 / (1400 \cdot 3.14) \approx 30 \text{ mm}$$

Conclusion for Titan: ablator thickness is ~30 mm, PCM thickness is ~1.65 m. The hybrid scheme does not work in pure form due to the excessive PCM thickness.

Alternative for Titan: paraffin instead of salt

If paraffin is used (melting point 45–65 °C), its heat of fusion is 200 kJ/kg, density — 800 kg/m³.

$$h_{\text{paraffin}} = (1.96 \cdot 10^9) / (200,000 \cdot 800 \cdot 3.14) \approx 3.9 \text{ m}$$

This is even worse. **Conclusion:** pure PCM is unsuitable for Titan.

2.9.4. Comparative table: Earth, Mars, Titan

Parameter	Earth (Soyuz)	Mars	Titan
Atmosphere	N ₂ /O ₂	CO ₂ (95%)	N ₂ /CH ₄ (95%/5%)
Pressure	1 atm	0.006 atm	1.5 atm
Entry speed	7.5 km/s	6–7 km/s	6–7.3 km/s
Peak heat flux	1.2 MW/m ²	1–3 MW/m ²	2.5–3 MW/m ²
Radiative heating share	<5%	50–70%	80–95%
Ablator material	AVCOAT / glass-phenolic	PICA	AQ60
Ablator thickness	~13 mm	~55 mm	~30 mm
PCM thickness (salt)	~0.95 m	~0.79 m	~1.65 m
Hybrid works?	Partially	Partially	No
Landing probability	—	5/5	4/5

2.9.5. Prospects for landing on Mars and Titan

Mars — the main candidate

Why Mars:

- There is an atmosphere — can decelerate
- Gravity 0.38g — lighter than on Earth
- Water in the form of ice at the poles and mid-latitudes
- Proximity to Earth (6–9 months of flight)
- Most studied — rovers are operating, orbital stations exist

Problems:

- Atmosphere is 100 times thinner than Earth's
- Radiative heating — up to 50–70% of total
- Chemistry is non-equilibrium — simple formulas give errors up to 2 times

What is needed for the hybrid:

- Ablator: PICA (Phenolic Impregnated Carbon Ablator)
- PCM: Glauber's salt is suitable, but a barrier layer against CO₂ is needed
- Calculation: 3D-CFD with chemical kinetics

Titan — "the second home" of humanity

Why Titan:

- Dense nitrogen atmosphere — surface pressure 1.5 atm
- Radiation protection — the atmosphere protects against cosmic rays
- Organics — methane, ethane, possible prebiotic compounds
- You can walk without a spacesuit — pressure is almost like on Earth, only an oxygen apparatus and warm clothing are needed

Problems:

- Temperature –179 °C — extreme cold
- Far from Earth (Saturn)
- Weak sunlight (100 times weaker than on Earth)

What is needed for the hybrid:

- Ablator: AQ60 (as on Huygens)
- PCM: not salt (it will freeze), but paraffin with a low melting point
- Calculation: almost like on Earth, but accounting for low temperature

2.9.6. Conclusion on the section

The methodology developed in this work is applicable for preliminary estimates of thermal protection for Mars and Titan. However:

- **For Mars**, the hybrid scheme requires PICA instead of AVCOAT and accounting for radiative heating (50–70% of total). The PCM thickness (0.79 m) is still large, but already closer to realistic.
- **For Titan**, the hybrid scheme does not work in pure form: the PCM thickness (1.65 m) exceeds the dimensions of the vehicle. However, the dense

atmosphere and moderate heating make Titan promising for using other types of thermal protection (for example, reusable radiative shields).

Main conclusion: thermal protection is not a universal task, but a complex problem requiring an individual approach for each target.

Conclusion

3.1. Conclusions based on the research results

Based on the analysis performed, the following conclusions can be formulated:

1. **Pure PCM is unsuitable** for thermal protection of re-entry vehicles: its mass (4.45–5.65 t) and thickness (0.95–2.25 m) exceed practically achievable values for a small vehicle. The reason is not only the low specific heat of fusion (200–254 kJ/kg), but also the extremely low thermal conductivity: the transient calculation showed that in 300 s only 15–17.5 mm of the layer will melt, while the surface heats up to 1900 °C. This confirms the necessity of an external ablative layer.
2. **Ablative protection** remains the most mass-efficient, but its single-use nature is the main obstacle to creating reusable vehicles. In addition, it creates a technogenic load on the atmosphere through the ejection of ablation products.
3. **The hybrid scheme** (external ablator + internal PCM sublayer of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$) is the optimal compromise solution. It reduces the volume of replaceable material by 30–50% [2], provides the potential for reusability, and reduces the environmental footprint of space launches.
4. **Glauber's salt** is preferable to paraffin due to its higher volumetric energy density (2.4 times), but requires additional design solutions: a barrier layer against corrosion and a post-landing cooling system for solidification (melting point ~32 °C).
5. **The economic assessment confirmed the benefit of the hybrid scheme.** Even under conservative assumptions (underestimated ablator cost, additional PCM mass ~2 t), the hybrid is economically more advantageous by ~1.1 billion RUB over 10 flights. The main source of benefit is the reduction of costs for replacing the ablative shield after each flight.
6. **The increase in vehicle mass requires recalculation of the parachute system.** Adding a PCM sublayer increases the vehicle mass, leading to an increase in parachute diameter by 31% and parachute mass by 72%. This creates feedback and adds ~70 kg to the mass, but does not outweigh the economic benefit of the hybrid.
7. **The methodology is applicable to other planets, but requires adaptation.** The hybrid scheme "ablator + PCM sublayer", justified for Earth, can be extended to other targets. For Mars, replacing AVCOAT with PICA and accounting for radiative heating (50–70% of total) is required. For Titan, the hybrid scheme does not work in pure form due to the large PCM

thickness (1.65 m), but the dense nitrogen atmosphere makes it promising for other types of thermal protection. This shows that thermal protection is not a universal task, but a complex problem requiring an individual approach for each planet.

3.2. Degree of completion and prospects for the development of the topic

This work represents a preliminary engineering estimate performed on the basis of a simplified model assuming complete melting of the PCM layer. In reality, due to the limited thermal conductivity of PCM, only a near-surface layer will melt during descent, which requires a more detailed transient thermal calculation.

Promising directions for the development of the topic:

1. Refinement of the transient calculation accounting for the temperature dependence of material properties, PCM decomposition, and surface mass loss.
2. Numerical modeling of the hybrid scheme in software packages (ANSYS, COMSOL) for optimizing layer thicknesses.
3. Experimental verification of the hybrid system's performance under ground-based thermal testing conditions.
4. Study of alternative PCMs with higher thermal conductivity and melting point.
5. Development of the barrier layer design and post-landing cooling system for Glauber's salt.
6. Assessment of the environmental effect of introducing hybrid systems at the scale of growing launch numbers.
7. Refinement of the economic assessment using real market prices for ablative materials, PCMs and inter-flight maintenance.
8. Evaluation of the PCM sublayer resource under multiple melt-solidify cycles. Glauber's salt is prone to degradation: loss of crystallization water and stratification into anhydrous sulfate and water. Without stabilization, the salt resource is estimated at 10–15 flights; with a stabilizer and barrier layer — 30–50 flights. Experimental studies are required to refine the real resource under space descent conditions.

3.3. Possibilities of application in the space industry

The results of the work can be used:

1. At the early design stages of reusable re-entry vehicles to estimate the mass and dimensions of the PCM sublayer as part of combined thermal protection.
2. In the development of reusable return capsules for delivering cargo from the ISS or orbital stations.
3. In creating thermal protection systems for next-generation crewed spacecraft, where reducing the cost of inter-flight maintenance is a critical factor.

4. As a basis for further research on optimizing hybrid thermal protection systems using modern PCMs and nanomaterials.
5. To reduce environmental load — reducing the ablator volume in the hybrid scheme reduces the amount of ablation products ejected into the atmosphere, which is consistent with modern trends of sustainable space exploration and reduction of technogenic impact on the upper atmosphere [4].
6. The developed numerical modeling methodology can be used for preliminary assessment of thermal regimes of promising thermal protection systems.
7. For interplanetary missions. The developed methodology can be used as a basis for preliminary thermal protection assessment in the design of re-entry vehicles for Mars, Titan and other bodies of the Solar System.

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